MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LSMDU

This limited coordination military specification has been prepared by the Naval Sea Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-C-24643. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight)

(C) First - Copper conductor, uncoated (see table I for size). Coated copper may be used at manufacturer's option where required to provide free-stripping insulation.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

(C) Third - Ethylene propylene rubber or cross-linked polyethylene insulation or at the manufacturer's option a composite with a primary insulation consisting of ethylene propylene rubber or cross-linked polyethylene. The composite layer may be substituted for up to 10 percent of the wall thickness (see table I for wall thicknesses).

Fourth - Nineteen conductors, cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence to be consecutive, starting with no. 1 from center outward. Fillers shall be used to form a firm, well-rounded assembly.

Fifth - Binder tape applied helically with overlap.

Sixth - Cross-linked polyolefin jacket. (See table I for wall thickness.) Cable surface marking required.

(C) denotes changes.

AMSC N/A FSC 6145

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Ⓒ TABLE I. Details.

Military part no. M24643/5	Type and size	Conductor size AWG	Insulation thickness (min) (inch)	Cable jacket thickness (nominal) (inch)	Overall diameter (max) (inch)	Conductor resistance per 1000 feet (max) (ohms)	Insulation resistance per 1000 feet (min) (megohms)	Cold bending mandrel diameter (max) (inch)
-01UN	LSMDU-6	12(class B)	0.028	0.060	1.000	1.73	100	13
-02UN	LSMDU-14	9(class B)	.040	.060	1.395	0.868	100	19
-03UN	LSMDU-23	7(class B)	.052	.075	1.765	.598	100	23
-04UN	LSMDU-40	4(class C)	.052	.075	2.040	.273	90	27
-05UN	LSMDU-60	2(class D)	.052	.075	2.330	.172	75	30

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C, maximum..	(see table I)
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2500
Insulation resistance - megohms/1000 feet, minimum.....	(see table I)
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for limits of water leakage.....	No failure
Group B:	
Cold bending, cable - (see table I for mandrel diameter).....	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum.....	700
Elongation - percent, minimum.....	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	250
Permanence of printing (conductor) - cycles, minimum	50
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

(C) UNIT ORDERING LENGTHS:

1000 foot nominal length for sizes 6, 14 and 23.
500 foot nominal length for sizes 40 and 60.

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-N329-01)

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